

Motility-induced phase separation

When slowing down causes clustering

Self-propelled particles can accumulate without an attractive potential. The simplest feedback is kinetic: particles remain longer where they move slowly, and local crowding can reduce their speed further. A one-dimensional run-and-tumble model makes the instability particularly transparent.

1 Eliminating the directional imbalance

Let $R(x, t)$ and $L(x, t)$ be densities moving right and left at positive speed v . Each particle switches direction at rate $\alpha/2$, so the directional imbalance relaxes at rate α . Neglect translational Brownian diffusion, alignment, birth, death, and fluid-mediated interactions:

$$\partial_t R = -\partial_x(vR) - \frac{\alpha}{2}R + \frac{\alpha}{2}L, \quad (1)$$

$$\partial_t L = +\partial_x(vL) + \frac{\alpha}{2}R - \frac{\alpha}{2}L. \quad (2)$$

Writing $\rho = R + L$ and $p = R - L$ gives

$$\partial_t \rho = -\partial_x(vp), \quad \partial_t p = -\partial_x(v\rho) - \alpha p. \quad (3)$$

On times long compared with α^{-1} and lengths large compared with the persistence length v/α , adiabatic elimination gives

$$p \simeq -\frac{1}{\alpha}\partial_x(v\rho), \quad J = vp \simeq -\frac{v}{\alpha}\partial_x(v\rho). \quad (4)$$

For prescribed spatially varying $v(x)$, a zero-current steady state satisfies $v\rho = \text{constant}$, so $\rho \propto 1/v$. Motility affects a steady distribution directly, unlike a passive particle's mobility in thermal equilibrium with a fixed potential.

2 Density-dependent speed creates feedback

Now adopt a local mean-field constitutive law $v = v(\rho)$. Equation (4) becomes

$$J = -D_{\text{coll}}(\rho)\partial_x\rho, \quad D_{\text{coll}} = \frac{v^2}{\alpha} \left(1 + \rho \frac{v'}{v}\right). \quad (5)$$

For $\rho = \rho_0 + \delta\rho e^{st+iqx}$, linearization yields $s = -D_{\text{coll}}(\rho_0)q^2$. The homogeneous state is unstable to long wavelengths when

$$\boxed{\frac{d \ln v}{d \ln \rho} < -1.} \quad (6)$$

An excess of particles then reduces the outgoing flux enough to amplify itself. This is a spinodal criterion in the stated closure, not a complete phase diagram. Adding a constant translational diffusivity D_t adds D_t to D_{coll} and can suppress the instability.

3 A limited bridge to equilibrium phase separation

Introduce fixed positive reference scales ρ_* and v_* . The same deterministic flux can be written

$$J = -M(\rho)\partial_x\mu, \quad M(\rho) = \frac{\rho v(\rho)^2}{\alpha}, \quad \mu = \ln \frac{\rho}{\rho_*} + \ln \frac{v(\rho)}{v_*}. \quad (7)$$

For a speed that depends only on the local density, define

$$\mathcal{F}[\rho] = \int f(\rho) dx, \quad (8)$$

$$f(\rho) = \rho \left(\ln \frac{\rho}{\rho_*} - 1 \right) + \int_0^\rho \ln \frac{v(s)}{v_*} ds. \quad (9)$$

Then $\mu = \delta\mathcal{F}/\delta\rho$ and

$$f''(\rho) = \frac{1}{\rho} + \frac{v'}{v}, \quad \frac{d\mathcal{F}}{dt} = - \int M(\partial_x\mu)^2 dx \leq 0 \quad (10)$$

for periodic or no-flux boundaries. The spinodal is exactly the region $f'' < 0$. This mathematical Lyapunov functional provides an equilibrium-like description of the local closure; it is not automatically the physical free energy or entropy production of a swimming organism.

4 Why the short-wavelength physics matters

Negative diffusion alone makes the truncated equation unstable at arbitrarily large q , where the elimination leading to Eq. (4) is invalid. A simple illustrative regularization is

$$\mathcal{F} = \int \left[f(\rho) + \frac{\kappa}{2} (\partial_x \rho)^2 \right] dx, \quad \mu = f'(\rho) - \kappa \partial_x^2 \rho, \quad \kappa > 0. \quad (11)$$

The resulting conserved dynamics has dispersion

$$s(q) = -M(\rho_0)q^2 \left[f''(\rho_0) + \kappa q^2 \right]. \quad (12)$$

When $f'' < 0$, the fastest-growing linear mode has $q^2 = -f''/(2\kappa)$. This is an initial length scale, not necessarily the final domain size: conserved phase separation can continue to coarsen.

As a local example, $v = v_0(1 - \rho/\rho_{\max})$ gives an instability for $\rho > \rho_{\max}/2$ within $0 < \rho < \rho_{\max}$. Near vanishing speed, additional packing and persistence effects demand a better model. Coexistence densities cannot be inferred from this threshold alone.

Run-and-tumble bacteria motivate the kinetic picture [1]. In biological populations, chemotaxis, growth, adhesion, and hydrodynamics can also cause clustering. Moreover, gradient corrections in active systems need not derive from a functional: an equilibrium common-tangent construction is justified only when the required mapping actually holds [2].

5 Check the fast-variable elimination against exact kinetics

Before making v depend on density, take constant v and α on an infinite line. Differentiating the continuity equation and using the imbalance equation gives the telegraph equation

$$\partial_t^2 \rho + \alpha \partial_t \rho = v^2 \partial_x^2 \rho. \quad (13)$$

The term $\partial_t^2 \rho$ cannot be neglected immediately after a particle is prepared. The velocity autocorrelation of the symmetric two-state process is $\langle \dot{x}(t) \dot{x}(0) \rangle = v^2 e^{-\alpha|t|}$, since the direction-flip rate is $\alpha/2$. Integrating the autocorrelation twice yields

$$\langle [x(t) - x(0)]^2 \rangle = 2v^2 \int_0^t (t-s) e^{-\alpha s} ds \quad (14)$$

$$= \frac{2v^2}{\alpha} \left[t - \frac{1 - e^{-\alpha t}}{\alpha} \right]. \quad (15)$$

At $\alpha t \ll 1$ this is $v^2 t^2$, the ballistic regime. At $\alpha t \gg 1$ it is $2(v^2/\alpha)t$ plus a constant, so the effective one-dimensional diffusivity is v^2/α . Defining the flip rate as α instead would change the relaxation rate by two; the rate convention must accompany a quoted diffusivity.

For density-dependent $v(\rho)$, linearize the two original fields around a uniform state with $p = 0$. Writing $v_0 = v(\rho_0)$ only in this calculation,

$$s \begin{pmatrix} \delta \rho \\ \delta p \end{pmatrix} = \begin{pmatrix} 0 & -iqv_0 \\ -iq(v_0 + \rho_0 v'_0) & -\alpha \end{pmatrix} \begin{pmatrix} \delta \rho \\ \delta p \end{pmatrix}. \quad (16)$$

Consequently,

$$s_{\pm} = \frac{-\alpha \pm \sqrt{\alpha^2 - 4v_0(v_0 + \rho_0 v'_0)q^2}}{2}. \quad (17)$$

Expanding the slow root for small q gives

$$s_+ = -\frac{v_0(v_0 + \rho_0 v'_0)}{\alpha} q^2 + O(q^4), \quad (18)$$

which reproduces D_{coll} in the overview. The other eigenvalue approaches $-\alpha$: it is the eliminated orientation mode. The approximation requires the slow relaxation or growth rate to be much smaller than α .

If $v_0(v_0 + \rho_0 v'_0) < 0$, the full local two-field model still has growth increasing without bound at large q , although it grows as $|q|$ rather than q^2 . Retaining the imbalance therefore does not by itself supply a microscopic regularization. The response to crowding must have a finite spatial scale, or other microscopic physics must be retained.

What carries over to higher dimensions?

For isotropic orientations in d dimensions with orientational correlation time τ , the long-time active contribution is $D_{\text{active}} = v^2 \tau / d$. Rotational diffusion, tumbles, and alignment determine τ differently. The one-dimensional calculation isolates the feedback mechanism; it does not fix the prefactors for a two- or three-dimensional bacterial suspension.

6 Finite sensing range regularizes the instability

Suppose particles respond to a locally averaged density

$$\bar{\rho}(x) = \int K(x-y)\rho(y)dy, \quad \int K(y)dy = 1, \quad (19)$$

where K is symmetric and nonnegative, and replace the speed by $v(\bar{\rho})$. The long-time current is now $J = -(v/\alpha)\partial_x[v(\bar{\rho})\rho]$. For a Fourier perturbation, $\delta\bar{\rho} = \widehat{K}(q)\delta\rho$, so

$$s(q) = -\frac{v_0^2}{\alpha}q^2 \left[1 + \frac{\rho_0 v'_0}{v_0} \widehat{K}(q) \right]. \quad (20)$$

Large-scale density variations still generate the original feedback because $\widehat{K}(0) = 1$. Variations much shorter than the sensing range are averaged out.

For a Gaussian kernel of variance ℓ^2 , $\widehat{K} = e^{-q^2\ell^2/2}$. At a state with $\rho_0 v'_0/v_0 = -2$, the unstable band is $0 < q\ell < \sqrt{2\ln 2}$. The dimensionless dispersion is shown below. Equation (20) still assumes slow orientational dynamics have been eliminated; the whole unstable band is within that approximation if $\ell \gg v_0/\alpha$.

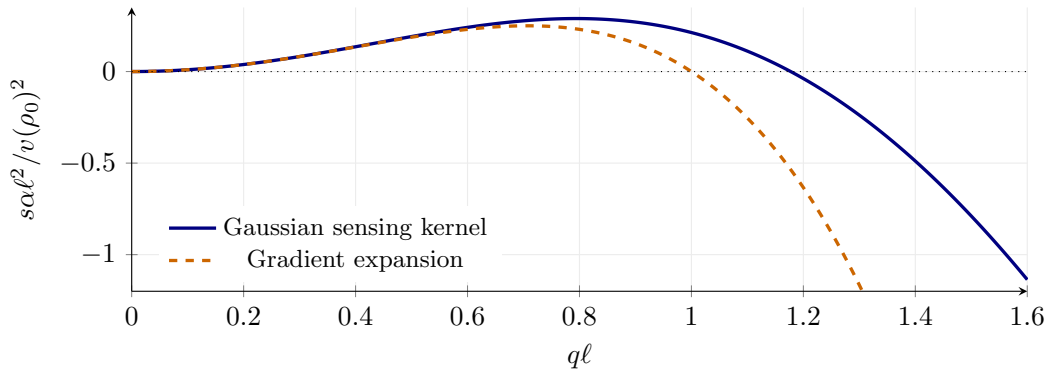


Figure 1: Analytic dispersion for Gaussian density sensing and its small- $q\ell$ expansion. Their disagreement near the band edge measures the limits of a gradient truncation.

Expanding the convolution gives $\bar{\rho} = \rho + (\ell^2/2)\rho_{xx} + \dots$. The effective chemical potential then has the form

$$\mu = \mu_0(\rho) - \kappa(\rho)\rho_{xx} + \dots, \quad \kappa(\rho) = -\frac{\ell^2}{2} \frac{v'(\rho)}{v(\rho)}. \quad (21)$$

Slowing with density produces positive κ . However, variation of an equilibrium-like gradient functional would instead give

$$\frac{\delta}{\delta\rho} \int \left[f(\rho) + \frac{\kappa(\rho)}{2} \rho_x^2 \right] dx = f'(\rho) - \kappa(\rho)\rho_{xx} - \frac{\kappa'(\rho)}{2} \rho_x^2. \quad (22)$$

The additional slope-squared term is generally absent from the kinetic expansion. This explicit mismatch shows why a local free-energy mapping need not survive at interfaces. Its absence cannot be repaired by declaring μ a thermodynamic chemical potential.

7 Spinodal, coexistence, and interface: three distinct calculations

When a gradient-flow mapping is justified, the spinodal condition $f'' < 0$ identifies local instability. Coexistence requires a separate constrained minimization. Let fractions θ and $1 - \theta$ occupy densities ρ_1 and ρ_2 , with fixed mean $\bar{\rho} = \theta\rho_1 + (1 - \theta)\rho_2$. Minimizing $\theta f(\rho_1) + (1 - \theta)f(\rho_2)$ under this constraint gives

$$f'(\rho_1) = f'(\rho_2) = \mu, \quad f(\rho_2) - f(\rho_1) = \mu(\rho_2 - \rho_1). \quad (23)$$

Thus the coexistence points share one tangent line. Between coexistence and spinodal, a uniform state can be locally stable but metastable to nucleation. For the active kinetic model with nonintegrable gradient terms, these equilibrium coexistence equations cannot simply be assumed.

A solvable local normal form

To see the distinction explicitly, set $\phi = \rho - \rho_c$ and use the illustrative Landau density

$$f(\phi) = \frac{a}{2}\phi^2 + \frac{b}{4}\phi^4, \quad a < 0, \quad b > 0. \quad (24)$$

This is a normal form near a symmetric mean-field critical point, not a derivation from the linear speed law above. Choose ρ_c large enough to keep both densities positive. Symmetry gives coexistence at $\phi = \pm\phi_0$, where $\phi_0 = \sqrt{-a/b}$ and $\mu = 0$. The spinodal is narrower:

$$f'' = a + 3b\phi^2 < 0 \iff |\phi| < \sqrt{\frac{-a}{3b}}. \quad (25)$$

For constant gradient coefficient κ , the stationary planar interface obeys $\kappa\phi_{xx} = a\phi + b\phi^3$. Multiplying by ϕ_x and integrating from either bulk phase yields

$$\frac{\kappa}{2}\phi_x^2 = f(\phi) - f(\phi_0) = \frac{b}{4}(\phi^2 - \phi_0^2)^2. \quad (26)$$

Separation of variables gives

$$\phi(x) = \phi_0 \tanh\left(\frac{x}{\sqrt{-2\kappa/a}}\right). \quad (27)$$

The interface width follows from κ and the bulk curvature; it is different from the fastest-growing wavelength of an initially uniform state. At $\phi = 0$, that linear wavelength is $2\pi\sqrt{2\kappa/|a|}$, while later evolution redistributes conserved mass and can coarsen beyond it.

Biophysical interpretation. A measured clustering threshold should be compared with the speed-density slope and persistence time under the same conditions. Check whether density sensing, steric slowing, or an attractive interaction is responsible. Continued growth or cell death changes the conservation law and may arrest coarsening. A deterministic mean-field instability in this one-dimensional toy model is not proof of a thermodynamic phase transition in a fluctuating one-dimensional system.

References

- [1] J. Tailleur and M. E. Cates, “Statistical Mechanics of Interacting Run-and-Tumble Bacteria” (2008). [arXiv:0803.1069](#).
- [2] M. E. Cates and J. Tailleur, “Motility-Induced Phase Separation,” *Annu. Rev. Condens. Matter Phys.* **6**, 219–244 (2015). [arXiv:1406.3533](#).